

22 **B** Resistors are connected in parallel, thus they have same p.d.

$$2I_1 = 3I_2 = 4I_3 \quad (1)$$

$$I_1 + I_2 + I_3 = I \text{ (current from power supply)} \quad (2)$$

$$\text{Sub (1) into (2): } (3/2)I_2 + I_2 + (3/4)I_2 = I$$

$$\text{So } I_2 = (4/13) I \quad (3)$$

$$\text{Sub (3) into (1) gives } I_1 = (6/13) I \text{ and } I_3 = (3/13) I$$

$$\text{So } I_1 : I_2 : I_3 = 6 : 4 : 3$$

